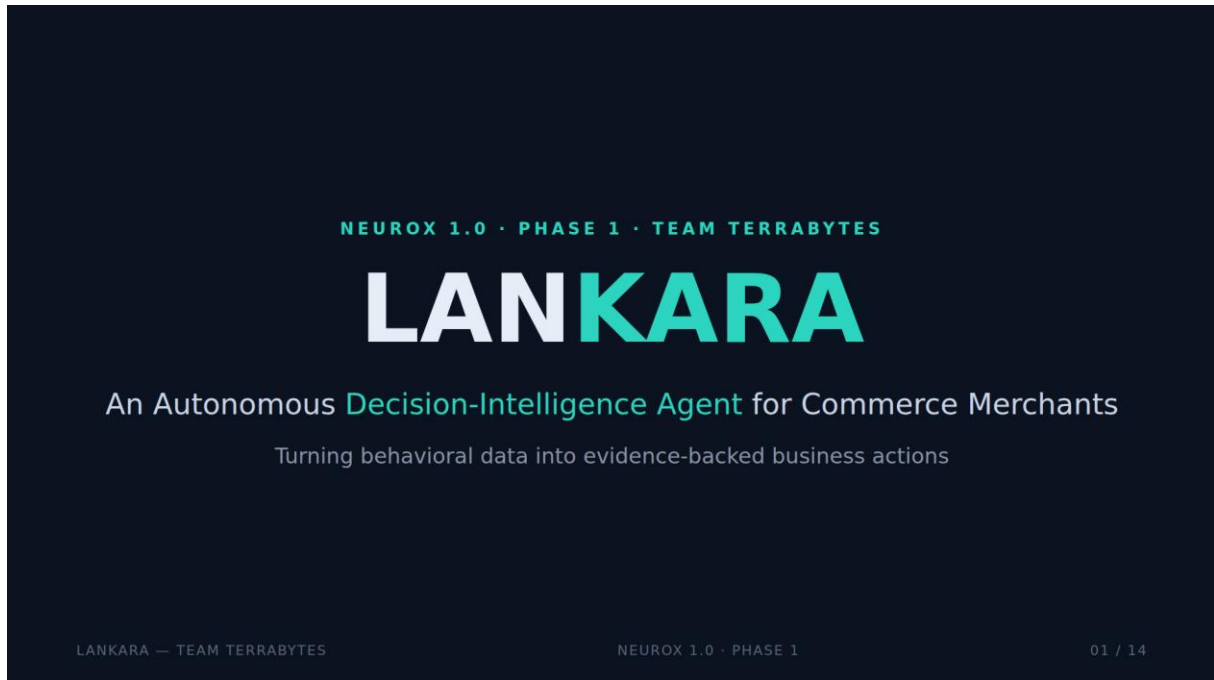




NEUROX 1.0 · PHASE 1 · DETAILED REPORT

Lankara

An Autonomous Decision-Intelligence Agent for B2B Commerce Operations

Turning merchant behavioral data into evidence-backed business actions — with a human always holding the final word.

TEAM TERRABYTES

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Executive Summary

Lankara is an autonomous decision-intelligence agent that sits inside a B2B commerce platform and does the judgment-intensive analytical work that ops, merchandising, and customer-success teams currently do by hand every week. Given a high-level goal — protect revenue, cut churn, optimize stock — the agent perceives the live behavioral event stream, plans which analytics tools to run, reasons over the combined evidence, and proposes a concrete, evidence-backed action. Nothing executes without a human decision at the Approval Queue.

This report expands on the five required sections of the NeuroX 1.0 Phase 1 submission — problem statement, value proposition, agentic loop design, architecture and tool mapping, and human-in-the-loop design — and adds supporting detail on data sourcing, the Phase 2 execution plan, evaluation methodology, and competition fit. The workflow sits squarely inside the in-scope domains of customer success and retention, financial operations and reporting, and procurement and vendor management, and is explicitly a B2B, non-consumer-facing system.

At a glance, Lankara replaces a five-step manual loop — export data, eyeball charts, guess at risk, draft actions by hand, wait for the next review cycle — with a continuously running agent that reasons over the same data merchants already collect, but does so every time the event stream shifts rather than once a week, and never moves money, inventory, or messages without explicit sign-off.

From dashboards that describe → to an agent that decides — with a human always holding the final word.

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1. Problem Statement

Mid-market and regional commerce businesses already collect rich behavioral and transactional data — page views, cart activity, purchases, returns, and support tickets — but a small operations, customer-success, and merchandising team still runs a weekly, judgment-intensive review loop entirely by hand.

1.1 The Manual B2B Decision Loop

The workflow we set out to solve repeats, almost unchanged, every single week across the merchants we spoke to while scoping this project:

- **Step 1** — Export sales and traffic data from dashboards or ad-hoc CSV dumps.
- **Step 2** — Eyeball charts for stockouts, slow movers, and weak categories.
- **Step 3** — Guess which accounts are slipping and which SKUs need promotion.
- **Step 4** — Manually draft reorders, discount rules, or retention outreach.
- **Step 5** — Wait days or weeks before the next review cycle can correct course.

This work is repetitive and slow, yet high-stakes: reordering, bundling, pricing, and retention calls are still made on intuition rather than on the evidence the platform already holds. Nobody on the team has the bandwidth to re-run this analysis continuously, so the loop only closes once a week at best — and every day inside that gap is a day of decisions made blind.

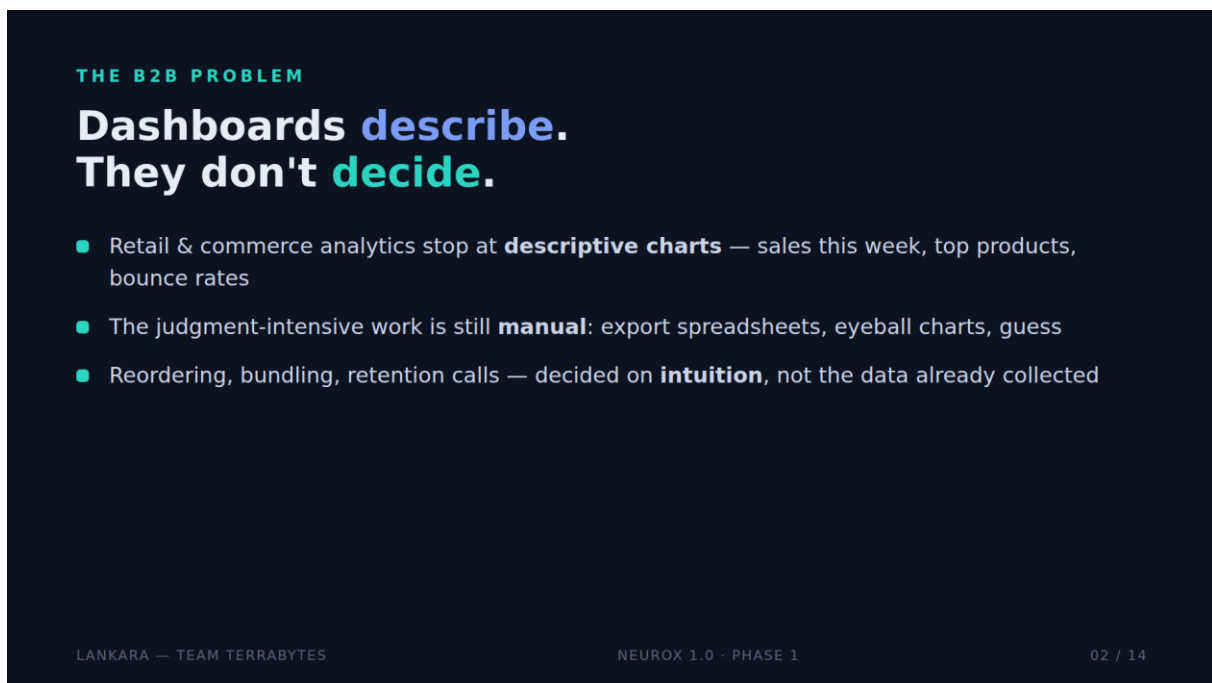


Figure 1. The B2B problem — dashboards describe, they don't decide.

1.2 The Cost of the Manual Loop

The concrete costs of running this loop by hand show up in three recurring places, all of which compound the longer the review cycle stays weekly instead of continuous:

Failure mode	What happens	Underlying cause
Overstocking	Working capital gets tied up in inventory nobody asked the data about.	Reorder decisions made on habit, not on forecast + demand signal.
Silent churn	At-risk customers go unnoticed until they have already left.	No continuous churn scoring against the live event stream.
Lagging prices	Pricing and promotion decisions trail the market by weeks.	Manual pricing review only happens on the weekly cadence.

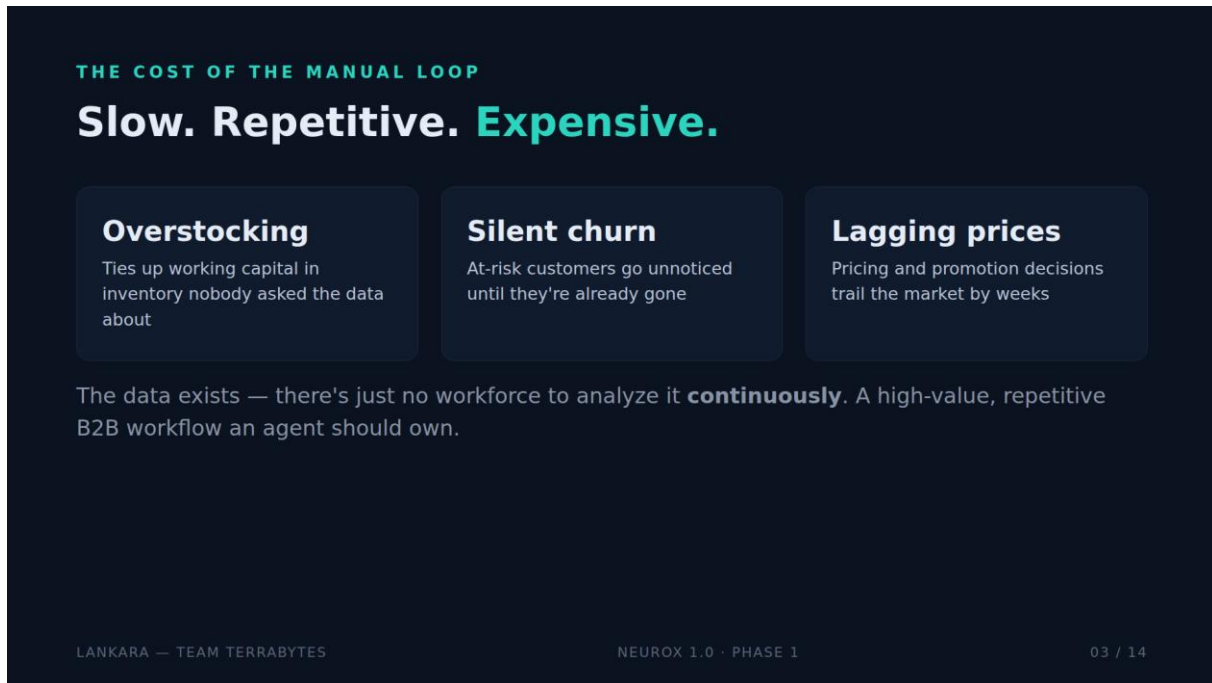


Figure 2. Slow, repetitive, and expensive — the manual loop's three recurring failure modes.

The underlying problem is not a lack of data. It is a lack of workforce to analyze that data continuously. This is precisely the profile of a high-value, repetitive B2B workflow that an autonomous agent should own: the inputs are already structured and machine-readable, the decisions are bounded and explainable, and the cost of delay is measurable in real currency.

1.3 Why This Is a B2B, In-Scope Workflow

Lankara operates entirely on the merchant side of the platform — it never talks to end consumers and never runs on hardware or IoT devices. The workflow maps cleanly onto three of the NeuroX in-scope domains:

- **Customer success and retention** — weekly churn scoring and retention-campaign drafting for at-risk merchant accounts and their customers.
- **Financial operations and reporting** — evidence packs quantify the working-capital and margin impact of every proposed action before it is approved.
- **Procurement and vendor management** — reorder proposals are demand-forecast driven purchase-order drafts routed to the merchant's inventory or ERP system.

No part of the system is a consumer-facing chat experience, a game, a patient-facing health tool, or a personal productivity utility — it is an operations copilot for the commerce team that runs the business.

2. Value Proposition

This problem is worth solving because every delayed or intuition-only decision burns either margin (the wrong discount), capital (the wrong stock level), or lifetime value (a retention call that came too late). Traditional BI software stops at description. Rule-based automation is brittle the moment the goal is ambiguous — a prompt like "sales look weak" has no single correct rule. An agentic workflow improves on both by reasoning over the goal, not just executing a fixed query.

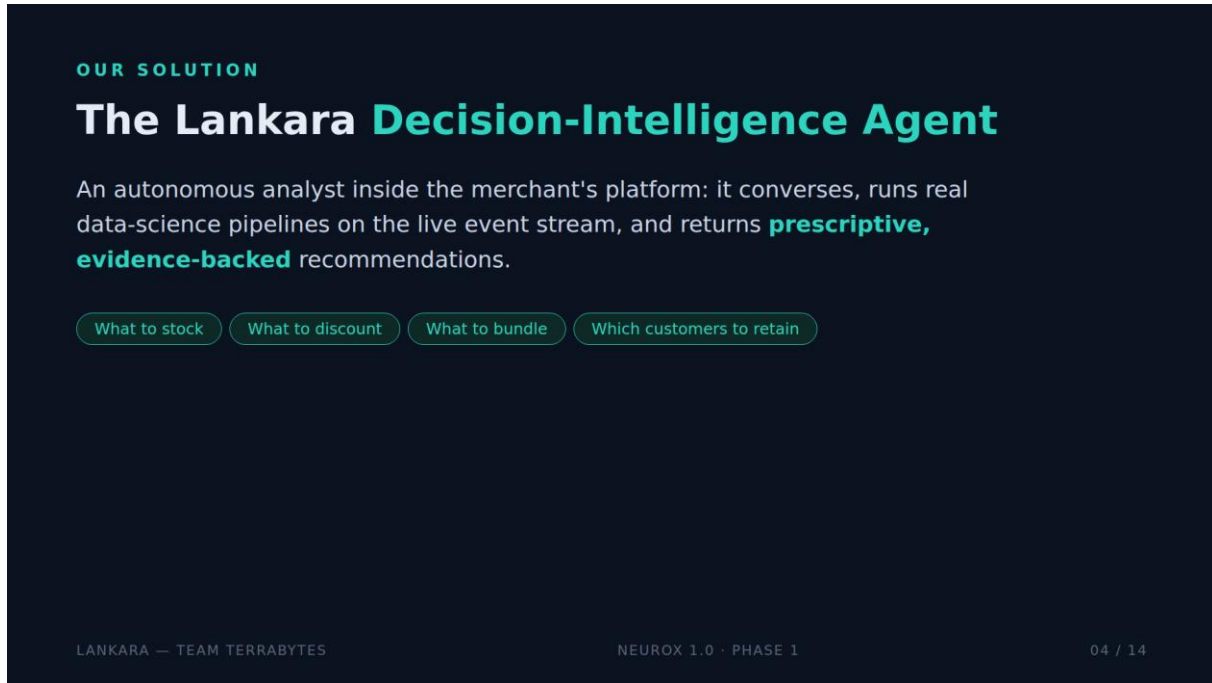


Figure 3. The Lankara Decision-Intelligence Agent — an autonomous analyst inside the merchant's platform.

2.1 How an Agent Improves on Current Practice

Dimension	Manual practice	Traditional BI / dashboards	Rule-based automation	Lankara agent
Analysis cadence	Weekly, when someone has time	On-demand, human-triggered	Fixed schedule	Continuous — re-runs whenever the event stream shifts
Tool selection	Whatever the analyst remembers to check	None — pre-built charts only	Hard-coded per rule	Chosen at runtime for the goal at hand
Handling ambiguity	Human interprets intuitively	Cannot interpret at all	Fails or misfires	Asks a clarifying question or scopes and states assumptions
Evidence	Verbal or absent	Chart only, no recommendation	None — silent execution	Chart + model confidence + exact

Dimension	Manual practice	Traditional BI / dashboards	Rule-based automation	Lankara agent
				data slice, every time
Control before action	Implicit in who does the work	N/A — no actions taken	None — executes blindly	Approval Queue — approve, edit, or reject
Failure behaviour	Analyst just tries again later	N/A	Breaks or produces bad output	Retries, falls back, degrades gracefully, never invents numbers

2.2 What the Agent Actually Answers

Framed from the merchant's point of view, Lankara is built to answer four recurring, high-stakes questions on demand, continuously, with evidence attached to every answer:

- **What to stock** — which SKUs to reorder now, and in what quantity, based on live demand forecasting.
- **What to discount** — which categories or SKUs are lagging the market and need a price or promotion adjustment.
- **What to bundle** — which products co-occur in baskets often enough to justify a bundle or cross-sell offer.
- **Which customers to retain** — which high-value accounts are showing early churn signals and need a retention touch.

2.3 Quantified Business Impact

Because every proposal in the Approval Queue carries its evidence pack, the business impact of adopting Lankara is measurable rather than anecdotal:

- **Working-capital efficiency** — continuous churn scoring surfaces at-risk high-value accounts while there is still time to intervene, rather than after the next invoice is missed.
- **Reduced overstock** — demand-forecast-driven reorder proposals reduce the gap between what is ordered and what actually sells.
- **Faster pricing response** — price and bundle proposals move on the same cadence as the market instead of trailing it by a full review cycle.
- **Compounding accuracy** — every action a human approves, edits, or rejects becomes feedback that tunes future proposals, so the agent's recommendations improve with use instead of drifting.

3. Proposed Agentic Loop

Lankara converts a high-level merchant goal into an evidence-backed action through dynamic reasoning, not a fixed script. The tool sequence is chosen at runtime by the agent itself, based on the goal and what the perception stage returns — it is not hard-coded per intent.

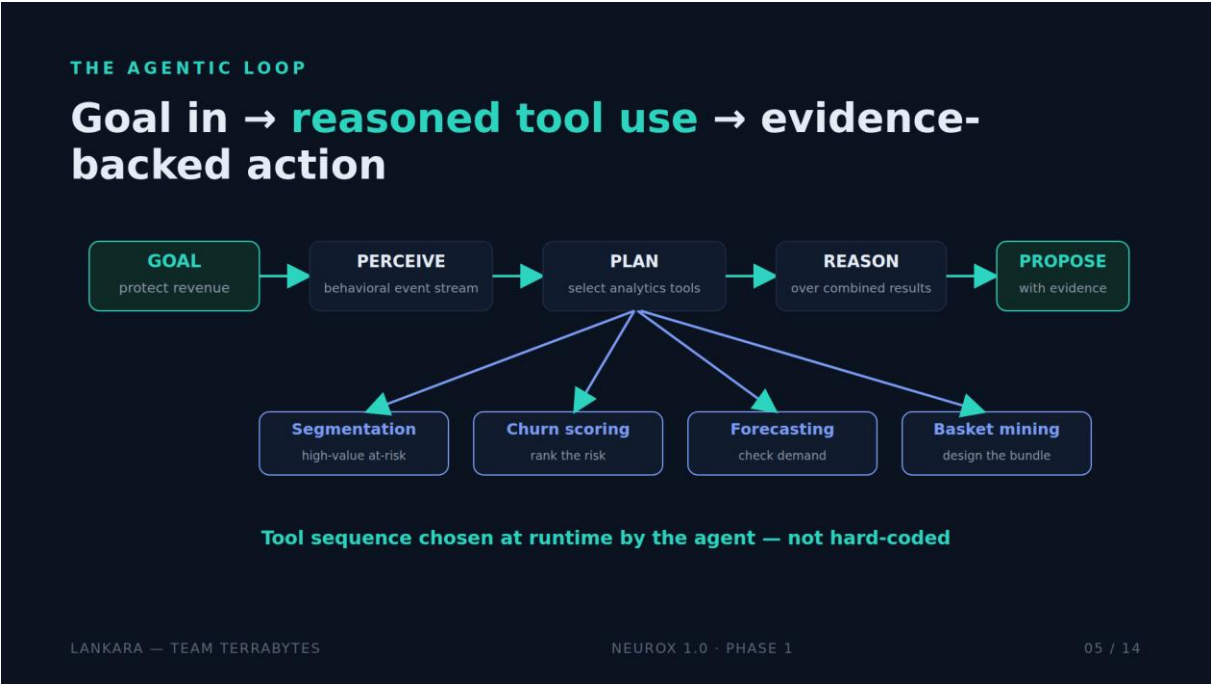


Figure 4. The agentic loop — goal in, reasoned tool use, evidence-backed action out.

3.1 Loop Stages in Detail

Stage	What happens	Example output
GOAL	A merchant or ops user states a high-level objective in natural language or selects one from a preset list.	"Protect revenue in the Kandy region this month."
PERCEIVE	The agent reads the live behavioral event stream, CRM snapshots, and order history relevant to the goal.	Last 30 days of views, carts, orders, and support tickets for Kandy-region accounts.
PLAN	The agent selects which analytics tools are needed for this specific goal — segmentation, churn scoring, forecasting, basket mining, or a combination.	Plan: run churn scoring first, then segmentation on flagged accounts.
REASON	Tool outputs are combined, conflicts are resolved, and a confidence score is attached to the resulting recommendation.	12 high-value accounts flagged at 78% churn risk; evidence attached.
PROPOSE	The agent emits a draft action plus its evidence pack into the Human-in-the-Loop Approval Queue.	Draft retention campaign for the 12 flagged accounts, awaiting approval.

3.2 Handling Ambiguous Inputs

If an ops user says something as underspecified as "sales look weak," the agent has two valid responses, and it chooses between them based on how much the ambiguity would change the analysis:

- **Ask** — if the region, category, channel, or time window materially changes the answer, the agent asks a single, targeted clarifying question before spending any tool calls.
- **Scope and state** — if a reasonable default scope exists, the agent scopes the analysis itself and states its assumption up front, so a human can redirect the plan before it goes further.

In both cases, assumptions remain visible in the evidence pack that eventually reaches the Approval Queue, so nothing the agent inferred is hidden from the human reviewer.

3.3 Recovering From Tool Errors

On tool failure or sparse data, the agent follows a strict degradation order rather than ever inventing a number:

- **1. Retry** — the agent retries the failed tool call once before treating it as unavailable.
- **2. Fall back** — the agent switches to an alternative model or a coarser data slice that can still answer the question.
- **3. Degrade** — if only partial evidence is available, the agent proceeds but reports a lower confidence score attached to the proposal.
- **4. Block, don't guess** — a hard outage produces a blocked proposal with a clear, human-readable failure reason rather than a silent skip or a guessed answer.

NOT A CHATBOT · NOT A SCRIPT

Built for **ambiguity** and **failure**

Ambiguous input — "sales look weak"

→ Asks a clarifying question, or

→ Scopes the analysis itself and **states its assumption** up front

Tool failure / sparse data

→ Retries, then switches to an alternative model

→ Degrades gracefully and **reports lower confidence** — never hallucinates

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Figure 5. Built for ambiguity and failure — not a chatbot, not a brittle script.

3.4 Worked Example

Consider a concrete run of the loop: an ops manager types "sales look weak in Kandy." The agent recognizes that "weak" is ambiguous without a comparison baseline and a category scope, so it scopes the analysis to all categories, compares the trailing 4 weeks against the prior 4, and states this assumption in its first response. It then plans a tool sequence — forecasting to check whether the dip is seasonal, followed by segmentation to see which customer tiers are driving it. Forecasting returns a healthy confidence score; segmentation flags a cluster of high-value accounts with rising cart-abandonment. The agent reasons over both outputs, concludes the dip is retention-driven rather than seasonal, and proposes a targeted discount plus a retention email for the affected segment — with the underlying chart, the model's confidence, and the exact account list attached as evidence. Nothing is sent until a human approves it in the queue.

4. Architecture & Tool Mapping

The system is modular and failure-tolerant by design. A LangGraph orchestrator functions as the agent core and tool router; every capability the agent can call runs as an independent microservice, so a failure in any one tool never takes the reasoning loop down with it.

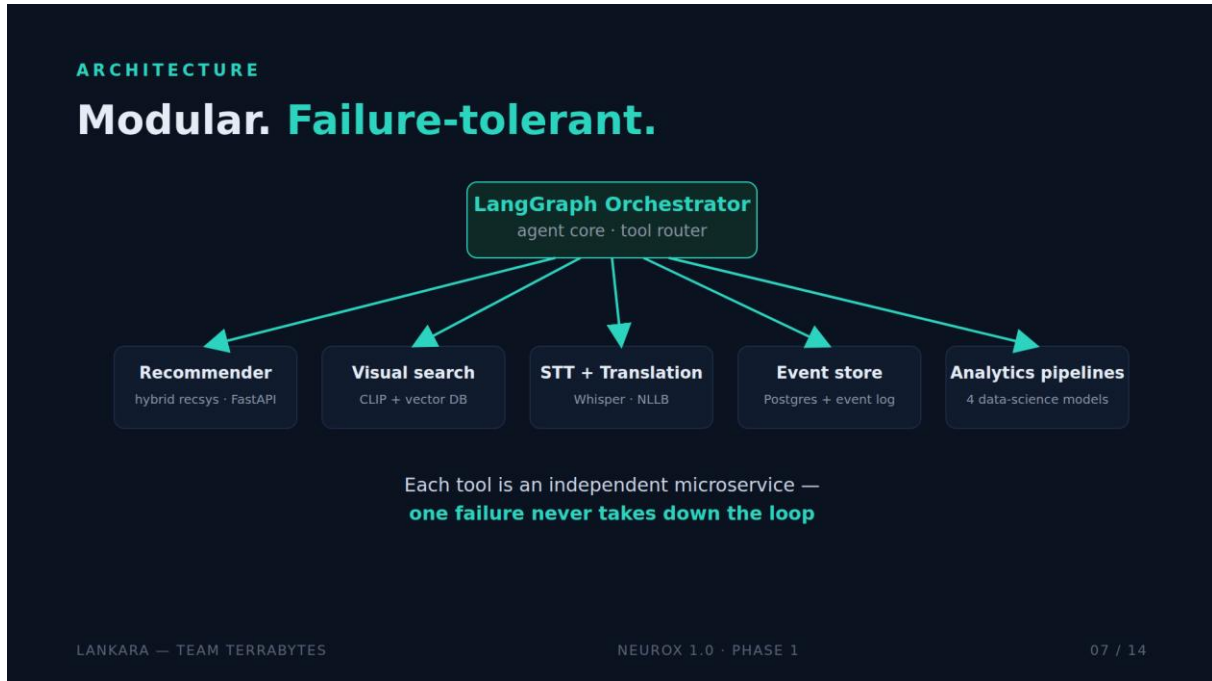


Figure 6. Modular, failure-tolerant architecture — each tool is an independent microservice.

4.1 System Layers

- **Presentation layer** — the merchant-facing conversational storefront and the ops-facing Approval Queue interface.
- **Agent core** — the LangGraph orchestrator that perceives, plans, reasons, and proposes.
- **Service layer** — the recommender, visual search, speech-to-text and translation, and analytics-pipeline microservices the agent calls.
- **Data layer** — the Postgres event store holding the live behavioral stream every service reads from and writes to.
- **ML / ops layer** — model registries, retries, and confidence scoring shared across every analytics pipeline.

4.2 External Tools the Agent Dynamically Selects

NeuroX 1.0 requires at least three external tools, APIs, or data sources the agent selects among at runtime. Lankara integrates five, all genuinely optional at plan time — the agent decides which to call based on the goal, not a fixed pipeline:

Tool / service	Role in the agent's reasoning
Analytics pipelines API	Segmentation (RFM + clustering), churn scoring (gradient boosting), demand forecasting (SARIMA / LightGBM), and basket mining (FP-Growth).
Hybrid recommender service (FastAPI)	Collaborative + content + session-based signals for bundle and cross-sell proposals.
Vector database (CLIP embeddings)	Visual product search over the catalog, used when a proposal needs to identify visually similar or substitute SKUs.
Whisper STT + translation (NLLB)	Voice and multilingual input into the platform, so ops staff can speak a goal in Sinhala, Tamil, or English.
Postgres event store	The live behavioral stream the agent perceives at the start of every loop — views, carts, orders, returns, tickets.

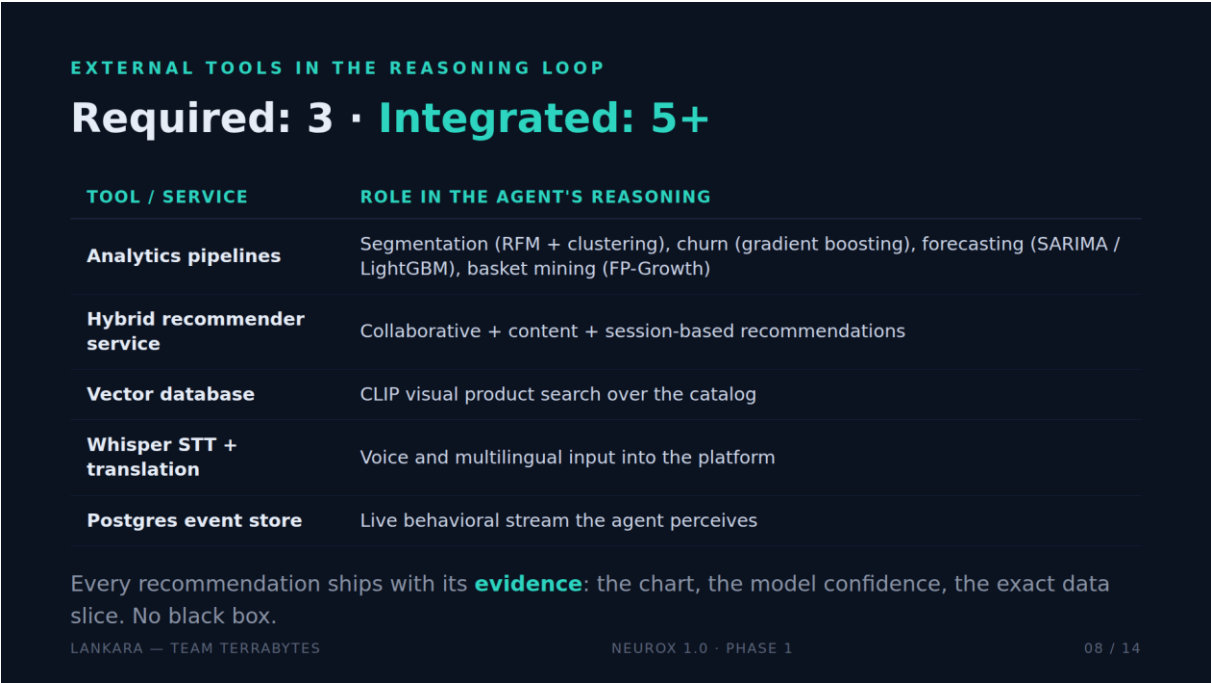


Figure 7. Required: 3 · Integrated: 5+ — every recommendation ships with its evidence.

4.3 Data Flow and Failure Tolerance

Every recommendation ships with its evidence: the chart, the model's confidence score, and the exact data slice it was computed from. There is no black box between a proposal and the numbers behind it. Because each tool is an independent microservice behind the orchestrator, a failure in, say, the recommender service does not stop the churn-scoring pipeline from completing — the orchestrator simply notes the missing input and lowers its confidence in any proposal that would have depended on it.

5. Human-in-the-Loop (HITL) Design

HITL in Lankara is a genuine control point, not a cosmetic confirm button bolted on after the fact. Any action that changes price, moves inventory, or reaches out to a customer must pass through the Approval Queue before it executes.

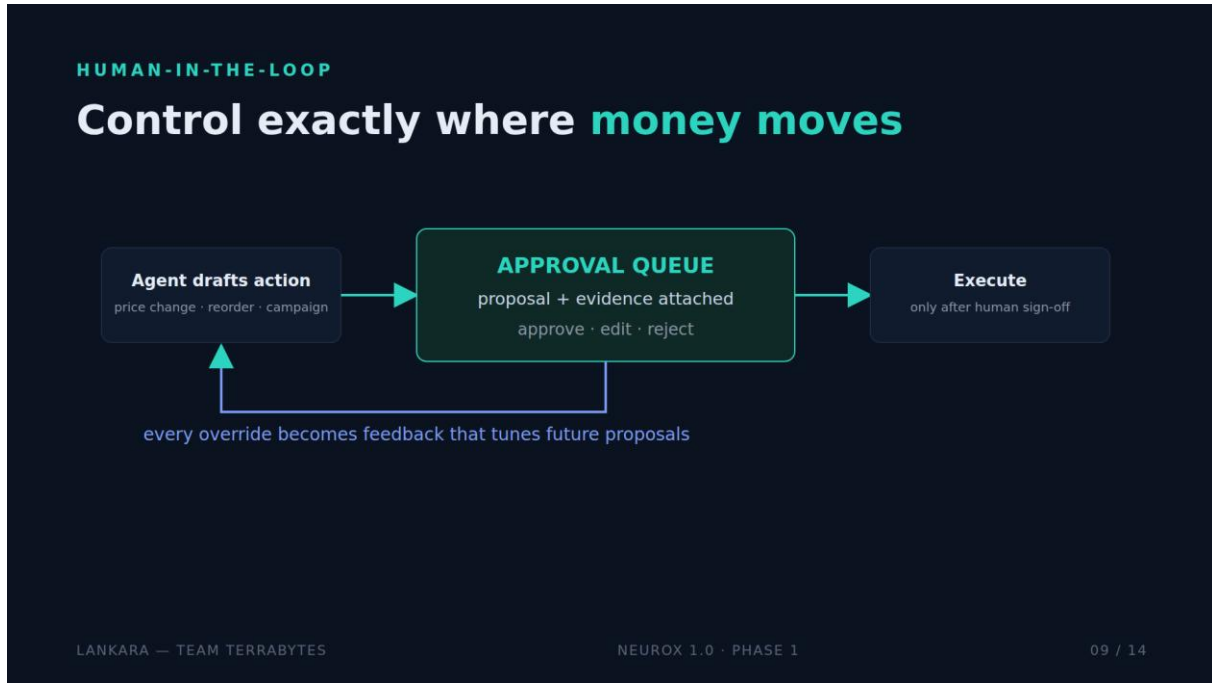


Figure 8. Control exactly where money moves — the Approval Queue.

5.1 The Approval Workflow

- **Draft** — the agent drafts the action — a price change, a reorder purchase order, or a retention campaign — with its full evidence pack attached.
- **Review** — a human reviews the proposal in the Approval Queue and can approve, edit, or reject it.
- **Execute** — the action executes only after explicit human sign-off — never before.

5.2 What Redirection Looks Like

Editing a proposal is a first-class action, not a workaround. Typical redirects a human reviewer can make without rejecting the proposal outright include:

- **Change segment** — narrowing or widening which customer segment a campaign targets.
- **Shrink discount** — reducing a proposed discount before it goes out.
- **Delay send** — holding a campaign or reorder for a later date.
- **Swap SKU list** — adjusting which products are included in a reorder or bundle proposal.
- **Block a channel** — preventing a specific channel (SMS, email, in-app) from being used for a given action.

5.3 Governance Table

Action type	Approval required	Typical reviewer
Price / discount change	Yes — always	Merchandising lead
Inventory reorder (PO draft)	Yes — always	Procurement / ops manager
Retention campaign send	Yes — always	Customer success lead
Internal analytics report	No — informational only	n/a

5.4 Feedback as a Learning Signal

Every override a human makes — an edit, a rejection, a delay — becomes feedback that tunes future proposals through a preference-memory and few-shot correction mechanism. If a reviewer consistently shrinks discount proposals for a given segment, the agent's next proposal for that segment starts closer to what the human actually wants, without ever being allowed to skip the review step itself.

6. Data Sources — The Storefront as Sensor Network

The agent's reasoning is only as good as the behavioral signal it perceives. Rather than treating data collection as a separate concern, the storefront layer that merchants run on the platform is designed to double as the sensor network that feeds the agent continuously.

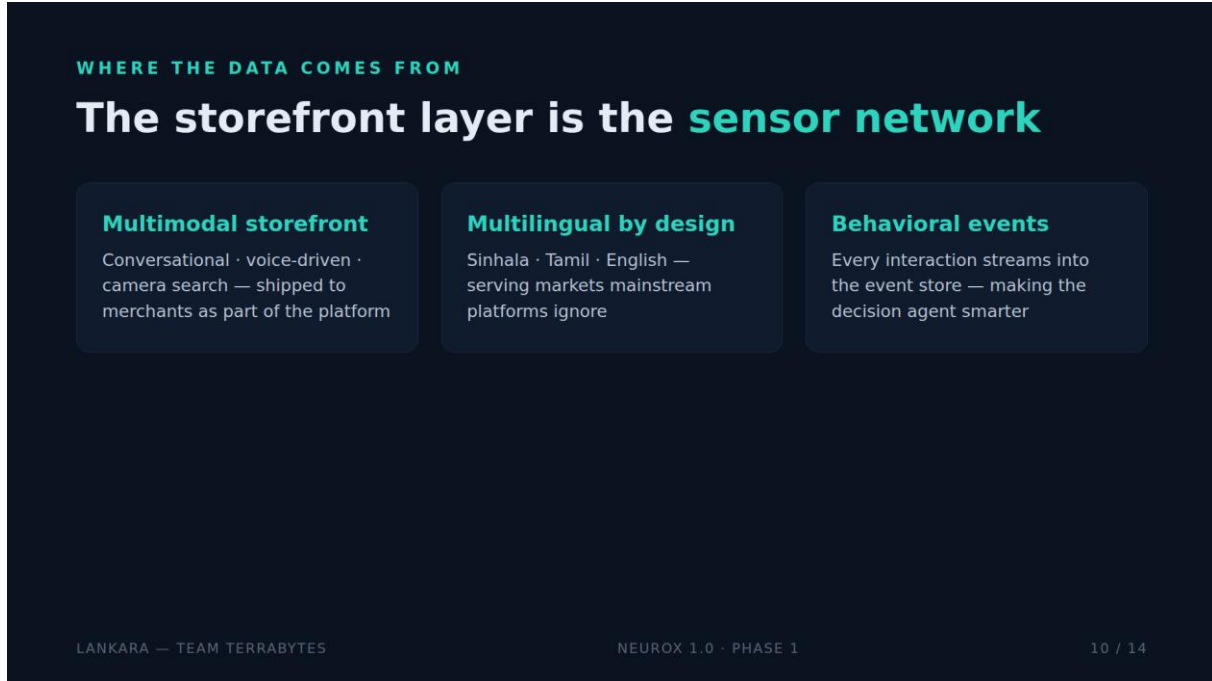


Figure 9. The storefront layer is the sensor network.

6.1 Multimodal Storefront Signals

The storefront supports conversational, voice-driven, and camera-based product search, shipped to merchants as part of the platform rather than as a bolt-on. Every one of those interaction modes generates structured events the agent can later reason over — a camera search for a product is as much a demand signal as a completed purchase.

6.2 Multilingual by Design

The storefront and the ops-facing agent both operate in Sinhala, Tamil, and English — serving the markets that mainstream commerce platforms in the region routinely ignore. This is not a translation layer bolted on after the fact; STT and translation are first-class tools the agent can call whenever a goal, a review, or a customer interaction arrives in any of the three languages.

6.3 Behavioral Event Schema

Event type	Captured detail	Feeds into
view	SKU, category, session, timestamp, device	Segmentation, forecasting
cart_add / cart_remove	SKU, quantity, session	Basket mining, forecasting
order	SKU list, price, discount applied, channel	Forecasting, recommender

Event type	Captured detail	Feeds into
return	SKU, reason code, order reference	Churn scoring, quality signal
support_ticket	Category, sentiment, account	Churn scoring
voice / visual search	Query embedding, matched SKUs	Recommender, catalog gaps

Every interaction streams into the Postgres event store, making the decision agent smarter with each additional session rather than requiring a periodic re-import of exported data.

7. Phase 2 — Technical Execution Plan

Phase 2 is de-risked with public datasets that stand in for the merchant event stream while the storefront's own data accumulates, and is built in four sprints, each shipping a runnable slice of the system rather than a demo shell.

PHASE 2 — TECHNICAL EXECUTION

De-risked with public datasets, built in 4 sprints

DATASET	DRIVES
RetailRocket	Clickstream, churn features
Online Retail II	Segmentation, forecasting
Instacart	Basket mining
H&M	Recommender

- Sprint 1** — Event store + analytics pipelines
- Sprint 2** — LangGraph agent + tool calling
- Sprint 3** — Approval-queue interface
- Sprint 4** — End-to-end evaluation

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Figure 10. De-risked with public datasets, built in four sprints.

7.1 Dataset-to-Capability Mapping

Dataset	Drives
RetailRocket	Clickstream features, churn labels
Online Retail II	Customer segmentation, demand forecasting
Instacart	Basket mining (FP-Growth) validation
H&M Personalized Fashion	Hybrid recommender training and evaluation

7.2 Sprint Plan

- **Sprint 1** — stand up the Postgres event store and the four core analytics pipelines against public datasets.
- **Sprint 2** — wire the LangGraph agent core and tool-calling logic to the analytics and recommender services.
- **Sprint 3** — build the Approval Queue interface — draft, review, approve/edit/reject, execute.
- **Sprint 4** — run an end-to-end evaluation across ambiguous-input handling, tool-failure recovery, and proposal quality.

This sequencing deliberately builds the reasoning loop and the human control point before optimizing any individual model — matching the review priorities of NeuroX 1.0 itself, which asks for design maturity before production polish.

8. Success Metrics & Evaluation

Every component of Lankara is measurable by design, both offline against held-out data and live in a demo run against the full reasoning loop.

MEASURABLE BY DESIGN

Every component has an **offline metric**

Recall@K
Recommendation quality

MAPE
Forecast accuracy

Precision
Churn prediction

Demo day: a goal in → tool calls reasoned through → an evidence-backed action awaiting **one human click.**

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Figure 11. Every component has an offline metric.

8.1 Offline Metrics by Component

Component	Metric	What it validates
Hybrid recommender	Recall@K	Whether the correct next product appears in the top-K recommendations.
Demand forecasting	MAPE	How close forecasted demand is to actual demand, SKU by SKU.
Churn scoring	Precision	Whether accounts flagged as at-risk are genuinely at risk, minimizing false alarms that would waste retention budget.
Basket mining	Support / confidence lift	Whether mined product associations are strong enough to justify a bundle.

8.2 Demo Day Acceptance Criteria

On demo day, the system is judged end to end on a single, unscripted run of the loop: a goal goes in, the agent reasons through tool calls it selects itself, and the result is an evidence-backed action awaiting exactly one human click. Success is not a canned trace — it is the agent handling whatever goal, ambiguity, or tool hiccup shows up live, in front of the judges, the same way it is designed to handle it in production.

9. Why NeuroX 1.0

Team Terrabytes found NeuroX 1.0 while exploring hackathon listings on Devpost, and it stood out immediately from other competition formats.

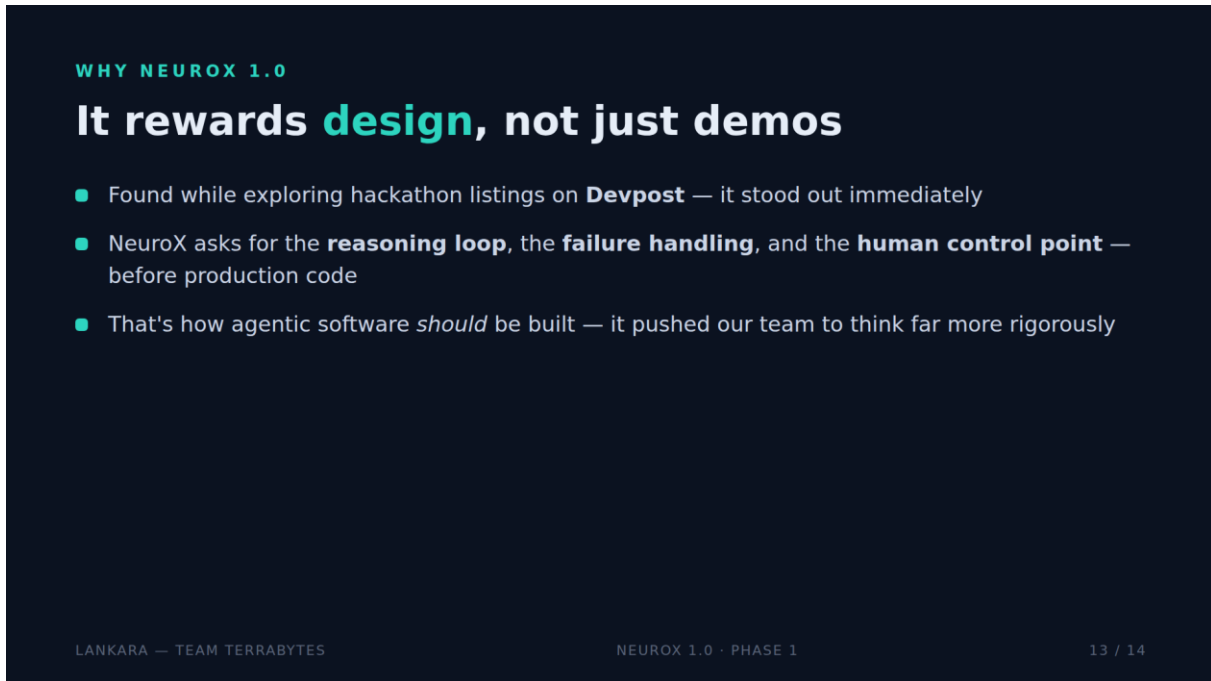


Figure 12. It rewards design, not just demos.

9.1 Fit and Rationale

- **Design-first scoring** — unlike most hackathon formats, NeuroX asks for the reasoning loop, the failure handling, and the human control point before any production code is written.
- **Real tool-use requirement** — the requirement for at least three dynamically selected external tools forced the team to design a genuinely modular architecture rather than a single monolithic pipeline.
- **HITL as a first-class requirement** — the mandatory Human-in-the-Loop control point matches how Team Terrabytes already believes agentic commerce software should be shipped — never autonomous over money without sign-off.

That's how agentic software should be built. The competition's own scoring rubric pushed the team to think far more rigorously about ambiguity handling and graceful degradation than a typical hackathon timeline would have allowed, and the result is a design the team would stand behind even outside the competition.

10. Risks & Mitigations

Risk	Impact	Mitigation
Sparse behavioral data for a new merchant	Low-confidence proposals, cold-start recommendations	Blend collaborative signal with content-based and session-based recommendation until enough history accumulates
Agent over-trusts a single tool's output	Biased or overconfident proposals	Reasoning stage cross-checks tool outputs against each other before a proposal is emitted; disagreement lowers confidence
Human reviewer fatigue in the Approval Queue	Rubber-stamp approvals that defeat the HITL control point	Evidence packs are ranked by materiality; low-impact proposals can be batch-reviewed, high-impact ones cannot
Tool or service outage mid-loop	Blocked or degraded proposals	Retry, fallback, and graceful degradation as described in Section 3.3 — never a silent skip
Multilingual input misclassified or mistranslated	Wrong scope assumed by the agent	Assumptions are always stated up front in the evidence pack, so a misread is caught before an action executes

11. Conclusion

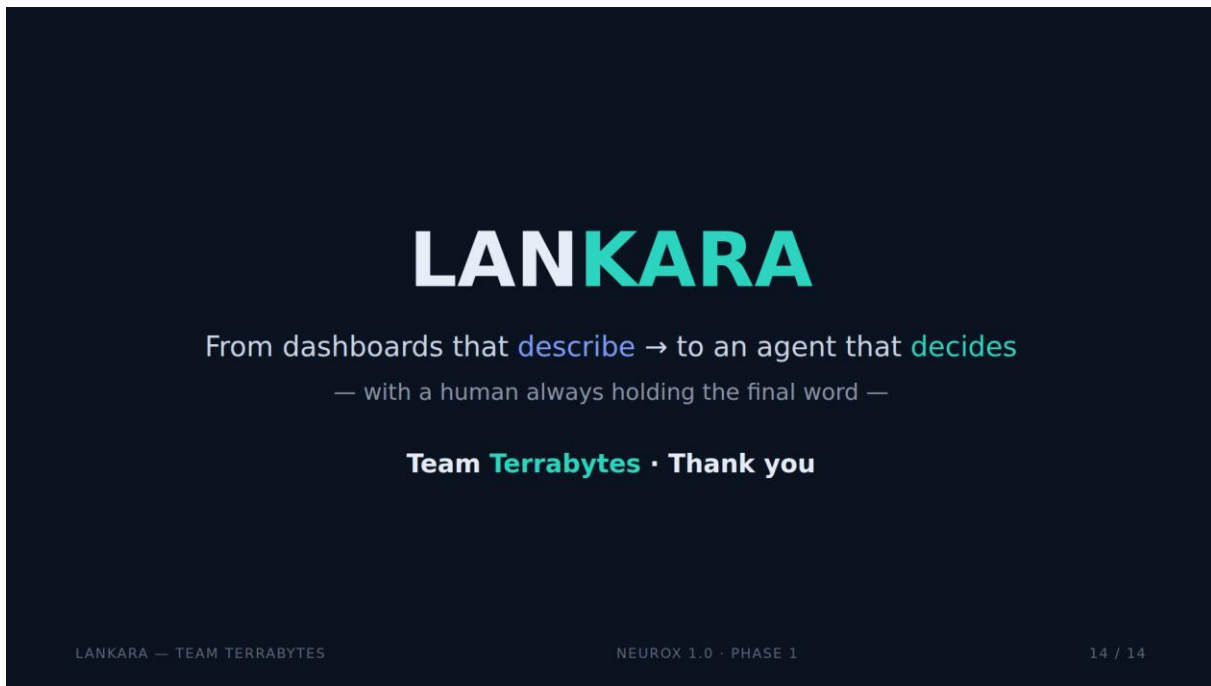


Figure 13. From dashboards that describe to an agent that decides.

Lankara takes a B2B workflow that is already high-value and already repetitive — the weekly reorder, pricing, and retention review — and hands it to an agent that runs continuously, chooses its own tools at runtime, and never moves money, inventory, or a customer message without a human decision. The design deliberately front-loads the parts that are hardest to get right and easiest to fake in a demo: ambiguity handling, tool-failure recovery, and a genuine approval gate.

Phase 2 de-risks the technical build against public datasets before the storefront's own event stream has accumulated enough history, and sequences the four-sprint plan so the reasoning loop and the human control point exist before any individual model is optimized. The result the team intends to demo is simple to state and hard to fake: a goal in, tool calls reasoned through by the agent itself, and an evidence-backed action awaiting one human click.

Lankara — Team Terrabytes: An Autonomous Decision-Intelligence Agent for B2B Commerce Operations.